THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 1

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A plant cell is placed in a solution and the cell contents shrink away from the wall. The solution is: 1

(a) pure water **(b)** hypotonic **(c)** hypertonic **(d)** isotonic

2. Which tissue would be most affected in a plant if the cambium is removed? 1

(a) water absorption **(b)** increase in length
(c) gas exchange **(d)** increase in girth

3. An organism is unicellular, eukaryotic and has chloroplasts and a flagellum. It belongs to: 1

(a) Monera **(b)** Plantae **(c)** Fungi **(d)** Protista

4. A farmer wants to grow a seedless banana plant. The method he uses is: 1

(a) vegetative propagation by suckers **(b)** cross-pollination
(c) grafting only **(d)** seed sowing

5. The distance from the Earth to the Sun is 1.5 x 10^11 m and the speed of light is 3 x 10^8 m/s. The time taken by sunlight is: 1

(a) 5000 s **(b)** 500 s **(c)** 5 s **(d)** 50 s

6. A cell has 8 chromosomes. After mitosis each daughter cell has: 1

(a) 8 **(b)** 2 **(c)** 16 **(d)** 4

7. A section of an animal tissue shows flat cells with a single layer. This is likely: 1

(a) bone **(b)** squamous epithelium
(c) cuboidal epithelium **(d)** columnar epithelium

8. A fish differs from an amphibian in having: 1

(a) a three-chambered heart **(b)** gills throughout life
(c) lungs **(d)** moist skin

9. A flower has stamens and a carpel. It is a: 1

(a) unisexual flower **(b)** female flower **(c)** bisexual flower **(d)** male flower

10. A 1 cm rise of the sea level due to melting ice would mostly affect: 1

(a) coastal low-lying areas **(b)** mountain tops
(c) deserts **(d)** polar caps only

11. The organelle which is found in all eukaryotic cells but whose number is highest in actively respiring cells: 1

(a) cell wall **(b)** vacuole **(c)** mitochondria **(d)** nucleus

12. A person can bend the arm at the elbow only in one plane. This joint is a: 1

(a) fixed **(b)** hinge joint **(c)** pivot joint **(d)** ball and socket

13. The correct sequence of taxonomic ranks for the tiger from the highest is: 1

(a) Animalia, Chordata, Mammalia, Carnivora, Felidae, Panthera, tigris

(b) Panthera, Felidae, Carnivora, Mammalia

(c) Chordata, Animalia, Mammalia, Carnivora

(d) Animalia, Mammalia, Chordata, Felidae, Carnivora, Panthera

14. The number of chromosomes in a human zygote is: 1

(a) 46 **(b)** 23 **(c)** 44 **(d)** 92

15. In the water cycle, the Sun provides energy for: 1

(a) condensation **(b)** evaporation **(c)** precipitation **(d)** run-off

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Why do red blood cells burst when placed in distilled water but not plant cells? 2

17. Why are the cells of the meristematic tissue closely packed with no intercellular spaces? 2

18. A student has a sample of a unicellular organism with a nucleus and cilia. In which kingdom will you place it and why? 2

19. Why do plants that are propagated vegetatively not show variation? 2

20. Calculate the time taken for sunlight to reach Earth: distance 1.5 x 10^11 m and speed 3 x 10^8 m/s, in minutes. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. A potato cube weighing 10 g is placed in distilled water and another in strong salt solution. Predict the masses after some time and explain. 3

22. Describe how the guard cells control the opening and closing of stomata. 3

23. Classify the following into the correct groups: banyan, fern, Spirogyra, moss, pine, rice. 3

24. Differentiate between the structure and functions of a sperm and an ovum. 3

OR

Explain the effect of albedo on the climate with the three surfaces: snow, forest, ocean, using values 0.85, 0.1, 0.06. 3

25. Explain two major differences between a bacterial cell and an animal cell with the genetic material in each. 3

OR

Explain how the tissue organisation is different in a multicellular plant and an animal with examples of conducting tissues. 3

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SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. Describe the structure and functions of the nucleus, including the nuclear membrane, nucleolus, chromatin and nuclear pores. 5

OR

A student cuts a thin section of a young stem and observes it under a microscope. Describe the steps and name the tissues seen from outside to inside, with the function of each. 5

27. A student collects ten organisms: mushroom, rose, whale, bacterium, Euglena, fern, snake, pigeon, earthworm and moss. Classify each into the kingdom and the group. 5

OR

Describe the sequence of events from the production of gametes to the birth of a baby in humans, with the organs involved. 5

END OF QUESTION PAPER — **Total: 50 Marks**

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